

CLINICAL DENTAL PRACTICE GUIDELINE: PERIODONTAL DIAGNOSIS, STAGING, AND GRADING

American Academy of Periodontology (AAP) and European Federation of Periodontology (EFP)

Guideline Code: DENT-PERIO-2024-V1

1. PERIODONTAL STAGING (SEVERITY & EXTENT OF DISEASE)

- Stage I (Initial Periodontitis):
 - * Clinical Attachment Loss (CAL): 1 to 2 mm at site of greatest loss.
 - * Radiographic Bone Loss (RBL): Coronal third (< 15% root length).
 - * Probing Depth (PD): Max <= 4 mm.
 - * Tooth Loss: No tooth loss due to periodontitis.
 - * Bone Loss Pattern: Mostly horizontal bone loss.
- Stage II (Moderate Periodontitis):
 - * CAL: 3 to 4 mm.
 - * RBL: Coronal third (15% to 33% root length).
 - * Probing Depth: Max <= 5 mm.
 - * Tooth Loss: No tooth loss due to periodontitis.
 - * Bone Loss Pattern: Mostly horizontal.
- Stage III (Severe Periodontitis with Potential for Additional Tooth Loss):
 - * CAL: >= 5 mm.
 - * RBL: Extending to middle third of root or beyond (> 33%).
 - * Probing Depth: >= 6 mm.
 - * Tooth Loss: <= 4 teeth lost due to periodontitis.
 - * Complexity: Vertical bone loss >= 3 mm, furcation involvement Class II or III, moderate ridge defect.
- Stage IV (Advanced Periodontitis with Potential for Dentition Loss):
 - * CAL: >= 5 mm.
 - * RBL: Extending to middle third of root or beyond.
 - * Tooth Loss: >= 5 teeth lost due to periodontitis.
 - * Complexity: Severe masticatory dysfunction, secondary occlusal trauma, tooth mobility degree >= 2, severe ridge defects, less than 20 remaining teeth (10 opposing pairs).

2. PERIODONTAL GRADING (RATE OF PROGRESSION & SYSTEMIC MODIFIERS)

- Grade A (Slow Rate of Progression):
 - * Objective Evidence: No bone loss over 5 years; RBL/age ratio < 0.25.
 - * Risk Factor Modifiers: Non-smoker, normoglycemic / non-diabetic.
- Grade B (Moderate Rate of Progression):

- * Objective Evidence: < 2 mm bone loss over 5 years; RBL/age ratio 0.25 to 1.0.
- * Risk Factor Modifiers: Smoker < 10 cigarettes/day; HbA1c < 7.0% in diabetic patients.
- Grade C (Rapid Rate of Progression):
- * Objective Evidence: $\geq$ 2 mm bone loss over 5 years; RBL/age ratio > 1.0.
- * Risk Factor Modifiers: Smoker $\geq$ 10 cigarettes/day; HbA1c $\geq$ 7.0% in diabetic patients.

3. TREATMENT PROTOCOLS & CLINICAL SEQUENCING

- Step 1: Periodontal health education, supragingival plaque control, chlorhexidine 0.12% rinse, smoking cessation counseling, glycemic control coordination.
- Step 2: Subgingival instrumentation (Scaling and Root Planing - SRP) using ultrasonic scalers and site-specific Gracey curettes under local anesthesia.
- Step 3 (if residual pockets $\geq$ 6 mm persist after 6-8 weeks re-evaluation): Periodontal flap surgery, resective surgery, or guided tissue regeneration (GTR) using bone grafts and barrier membranes.
- Step 4: Supportive Periodontal Therapy (SPT) / Maintenance visits every 3 months.